

Here is the table for **Discrete Distributions**.

Distribution	PMF $P(X = k)$	Mean $E[X]$	Variance $\text{Var}(X)$	MGF $\phi(t) = E[e^{tX}]$
Bernoulli (p)	$p^k(1-p)^{1-k}$	p	$p(1-p)$	$1-p+pe^t$
Binomial (n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$(1-p+pe^t)^n$
Poisson (λ)	$\frac{e^{-\lambda}\lambda^k}{k!}$	λ	λ	$e^{\lambda(e^t-1)}$
Geometric (p)	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Negative Binomial (r, p)	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Hypergeometric (N, K, n)	$\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	$n\frac{K}{N}$	$n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$	(No simple closed form)

Notation:

- k : The random variable value (e.g., number of successes).
- n : Number of trials (or sample size for Hypergeometric).
- p : Probability of success.
- r : Number of successes needed (for Negative Binomial).
- N : Total population size (Hypergeometric).
- K : Total successes in the population (Hypergeometric).

Here is the table for **Continuous Distributions**.

Distribution	PDF $f(x)$	Mean $E[X]$	Variance $\text{Var}(X)$	MGF $\phi(t) = E[e^{tX}]$
Uniform (a, b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Exponential (λ)	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Normal (μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$
Gamma (α, λ)	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$
Chi-Squared (n)	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{\frac{n}{2}-1}e^{-\frac{x}{2}}$	n	$2n$	$(1-2t)^{-n/2}$
Beta (α, β)	$\frac{1}{B(\alpha, \beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	(No simple closed form)
Cauchy (x_0, γ)	$\frac{1}{\pi\gamma[1+(\frac{x-x_0}{\gamma})^2]}$	Undefined	Undefined	Does not exist

Notation:

- $\Gamma(\alpha)$: The Gamma function, $\Gamma(n) = (n-1)!$.
- $B(\alpha, \beta)$: The Beta function, a normalizing constant.
- t : MGF parameter (valid only where the integral converges, e.g., $t < \lambda$ for Exponential).